LGM: A Calibrated, Certifiable Factorized Neural Hawkes Process for Limit-Order-Book Simulation

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Abstract

Neural marked temporal point processes (MTPPs) predict the next limit-order-book (LOB) event well but *simulate* order flow poorly: in free roll-out they over-disperse or explode, and their stability cannot be honestly certified. We introduce LGM (Linear Ground $\times$ soft-max Marks), which factorizes the per-type intensity into a *linear* scalar ground rate and a *deep* soft-max mark head. Because the marks live on the probability simplex they are rate-neutral, so the total rate remains a pure linear Hawkes process regardless of mark-net nonlinearity. This preserves the exact stationary-mean formula, which we *pin* analytically to solve the train/simulate calibration mismatch, and yields a gauge-free branching ratio as a closed-form stability certificate. On Gemini ETH–USD data (62 event types), LGM is simultaneously calibrated (free-roll rate within $\sim$7% of real), a competitive predictor, clustered (correct Fano profile), and certifiable—the first model in our study to achieve all of these together.

Introduction

Order-book simulation underpins market-making research, backtesting, and the reinforcement learning of execution policies (Avellaneda and Stoikov 2008). Self- and mutually-exciting Hawkes processes are the classical model of clustered order flow (Hawkes 1971; Bacry, Mastromatteo, and Muzy 2015), and neural MTPPs extend them with expressive state (Mei and Eisner 2017). Yet a model that predicts the next event accurately can still fail as a *simulator*: under free roll-out the event rate runs away or collapses, and the stability of the recurrent dynamics cannot be read off the trained weights. We trace this to two causes—a train/simulate calibration mismatch and a gauge pathology in the rate read-out—and remove both by construction.

The LGM Factorization

The per-type intensity factorizes into a timing rate and a mark distribution,

$$\lambda_k(t) = \Lambda(t)\, p(k \mid t), \qquad \sum_k \lambda_k(t) = \Lambda(t), \tag{1}$$

where the ground rate is a multi-timescale *linear* Hawkes process and the marks are a deep soft-max,

$$\Lambda(t) = \mu_0 + \sum_m a_m\, s_m(t), \qquad p(\cdot \mid t) = \mathrm{softmax}\big(z_\theta(t)\big). \tag{2}$$

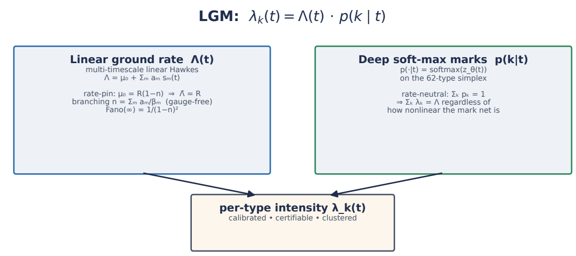

Figure 1: LGM factorizes the per-type intensity into a linear, rate-pinned ground rate $\Lambda(t)$ and a deep, rate-neutral soft-max over the 62 event types $p(k \mid t)$. Nonlinearity is confined to the simplex, where it cannot perturb the calibrated rate or the branching-ratio certificate.

Since the soft-max lies on the simplex, $\sum_k p_k = 1$, the marks are *rate-neutral*: the total rate Λ is unaffected by the mark-net nonlinearity. The stationary mean therefore survives in closed form,

$$\bar{\Lambda} = \frac{\mu_0}{1-n}, \qquad n = \sum_m \frac{a_m}{\beta_m}, \tag{3}$$

and we *pin* it by setting $\mu_0 = R_{\mathrm{target}}(1-n)$, so $\bar{\Lambda} = R_{\mathrm{target}}$ by construction. The branching ratio n is gauge-free and serves as a stability certificate, with $\mathrm{Fano}(\infty) = 1/(1-n)^2$.

Results

Table 1 reports the headline comparison on Gemini ETH–USD (62 event types, real rate $\approx$2.38 ev/s). LGM is uniquely calibrated *and* a competitive predictor *and* clustered *and* certifiable; the branching ratio is a single interpretable knob that sets the Fano-vs-scale curve via $1/(1-\rho)^2$.

Discussion

LGM removes the two failure modes of neural LOB simulators by construction: the analytic rate-pin (Eq. 3) fixes the train/simulate calibration mismatch without a stateful loader, and the direct linear ground read-out keeps the branching ratio gauge-free and measurable from weights. Honest caveats remain: return tails are $\sim$2$\times$ lighter than the robust empirical target, the model is mildly over-reflexive,

Table 1: Headline comparison on Gemini ETH–USD (62 event types; real rate $\approx$2.38 ev/s). GenAcc: genuine next-event accuracy.

Model	GenAcc	Fano(1s)	ρ	roll rate	sim status
Compound Hawkes	0.18	8.8	0.64	$\sim$2	stable, no mem.
s2p2 (predictor)	**0.32**	41	—	—	over-disperses
NMH / GMH	0.26	expl.	1300/0.8	52–100	explode / Pois.
LGM (n=0.86)	**0.29**	**6.0**	**0.86**	**2.22**	**calibrated**

and raw 1 s kurtosis/skew are outlier-dominated and should be read winsorized or at $\geq$5 s buckets. Action-conditioning toward a market-making world model is the natural next step.

References

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